CONCEPT PROPOSAL

Clean the Beach: Sri Lankan Coastal Cleanup Experience

SCENARIO / APPLICATION DOMAIN

This project is a first-person interactive digital environment based on a Sri Lankan coastal beach. The environment represents a polluted beach area where the player must collect plastic waste and help restore the cleanliness of the beach. The project connects with environmental awareness, coastal protection, and responsible human behavior in public natural spaces. It is designed as a short interactive awareness experience rather than a traditional entertainment game.

The player explores a tropical beach environment with sand, palm trees, rocks, beach props, water, and scattered trash items. The main goal is to collect trash before the timer ends while avoiding unsafe areas such as deep water or restricted zones. The experience encourages users to understand how small cleanup actions can improve the environment.

TARGET USER

The target users are students, young adults, and general players who are interested in simple first-person games and environmental awareness. The project is especially suitable for Sri Lankan users because it represents a familiar beach environment. The experience is designed to be easy to understand, visually clear, and engaging for users with basic game-playing skills.

INTELLIGENT BEHAVIOR

The system demonstrates intelligent behavior by reacting to the player's actions. When the player moves near a trash object, a pickup prompt appears. If the player presses E, the trash disappears, a sound plays, and the score increases. The game also detects unsafe areas and gives warning feedback when the player moves outside the safe zone.

PLANNED ADAPTIVE FEATURES

The environment adapts through visual, audio, and gameplay feedback. The score and timer update during gameplay, collection sounds play when trash is picked up, and beach ambience creates immersion. When the player enters an unsafe area, a warning message and alert sound are activated. These responses make the environment feel dynamic rather than static.

INTENDED USER EXPERIENCE

The intended experience is clear, simple, and immersive. The first-person view helps the player feel present inside the beach environment. The tutorial screen explains controls, while the UI shows the timer, score, pickup prompts, and warning messages. The player should easily understand the objective and receive feedback for every important action.

SELF-STUDY / INNOVATION DIRECTION

For the self-study component, this project explores an eco-feedback system. The system gives immediate feedback for environmentally positive actions, such as collecting trash, and negative actions, such as entering unsafe areas. This helps show how interactive media can encourage awareness, learning, and responsible behavior through direct user participation.

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